

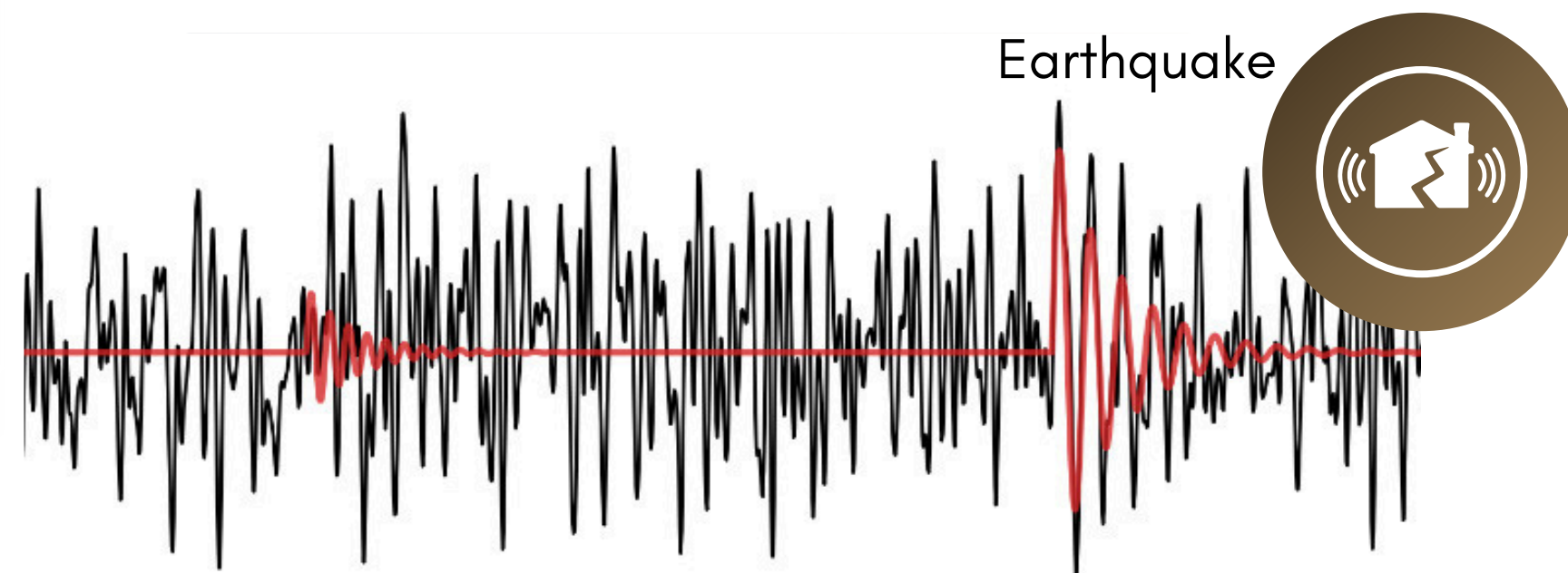
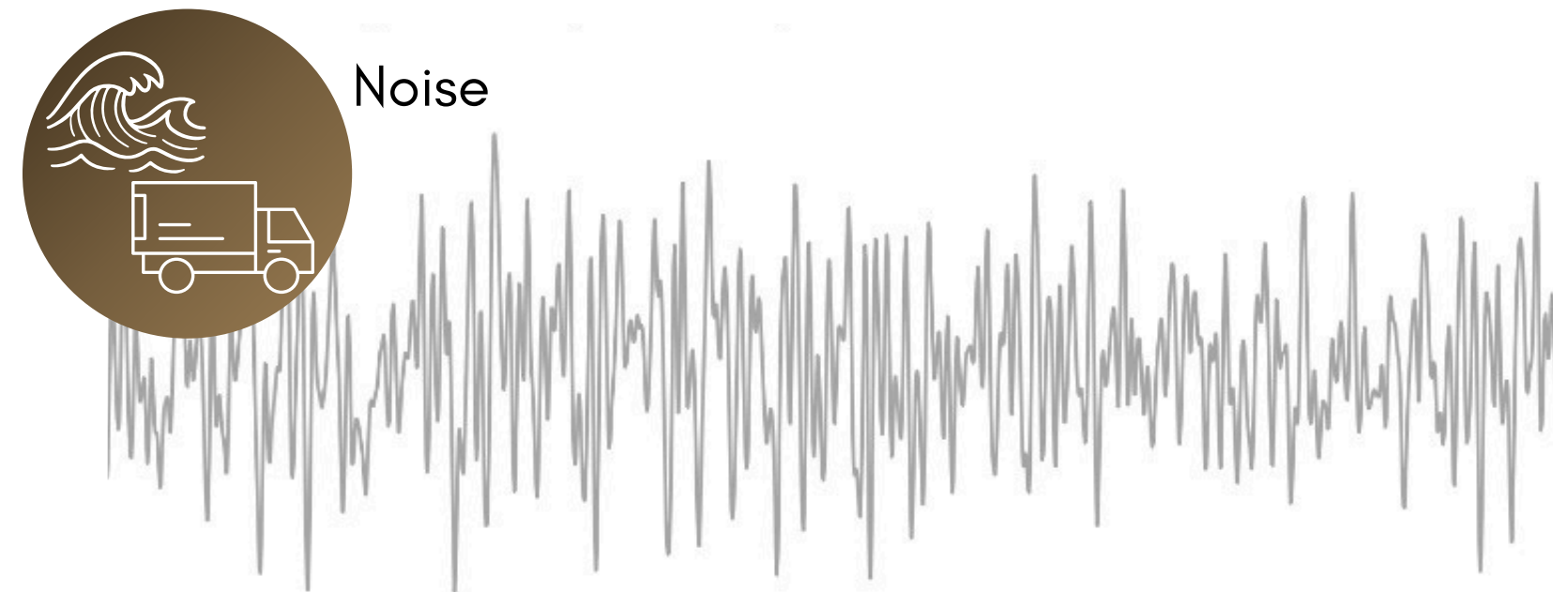
Cattaneo & Melchionna

D.E.E.P.

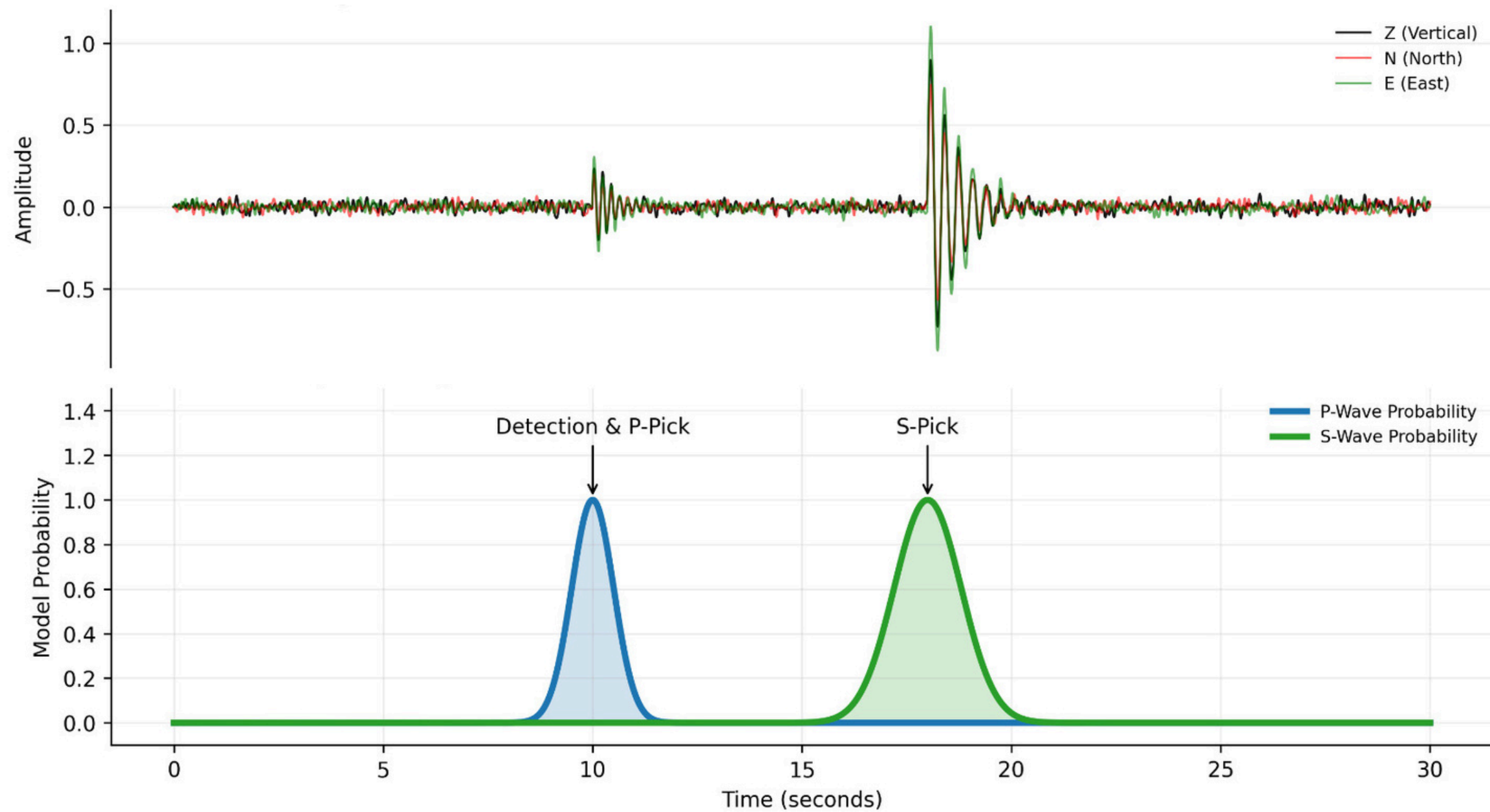
**Unified Deep Learning Model for Simultaneous
Seismic Event Detection and Phase Picking.**

D: DETECTION

- The Problem: Daily seismic data volume (Terabytes) > Human capacity.
- Task 1 (Detection): Binary classification (Is this an earthquake or just noise?).
- Data Source: INSTANCE Dataset.
- Strategy: Balanced training (50% Noise / 50% Earthquake) to teach the model to ignore background interference.

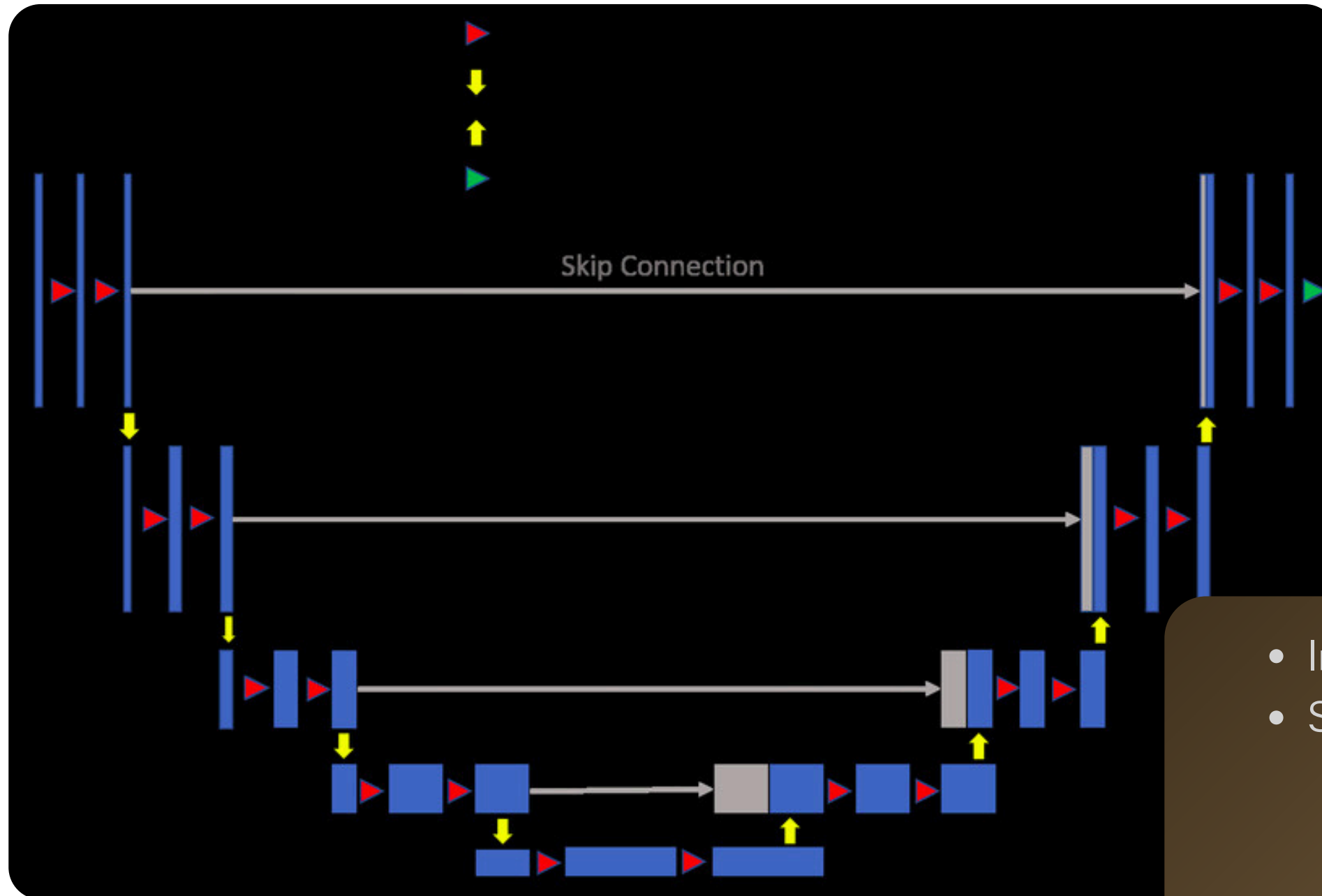


E: EXACT PHASING



- Task 2 (Picking): Identifying precise arrival times of Primary (P) and Secondary (S) waves.
- Why? The time difference between P and S is required to locate the earthquake's source.
- Method: Time-Series Segmentation.
- Target: Instead of predicting a single number, the model outputs "Probability Masks" (Gaussian distributions) over the arrival area.

E: ENGINE



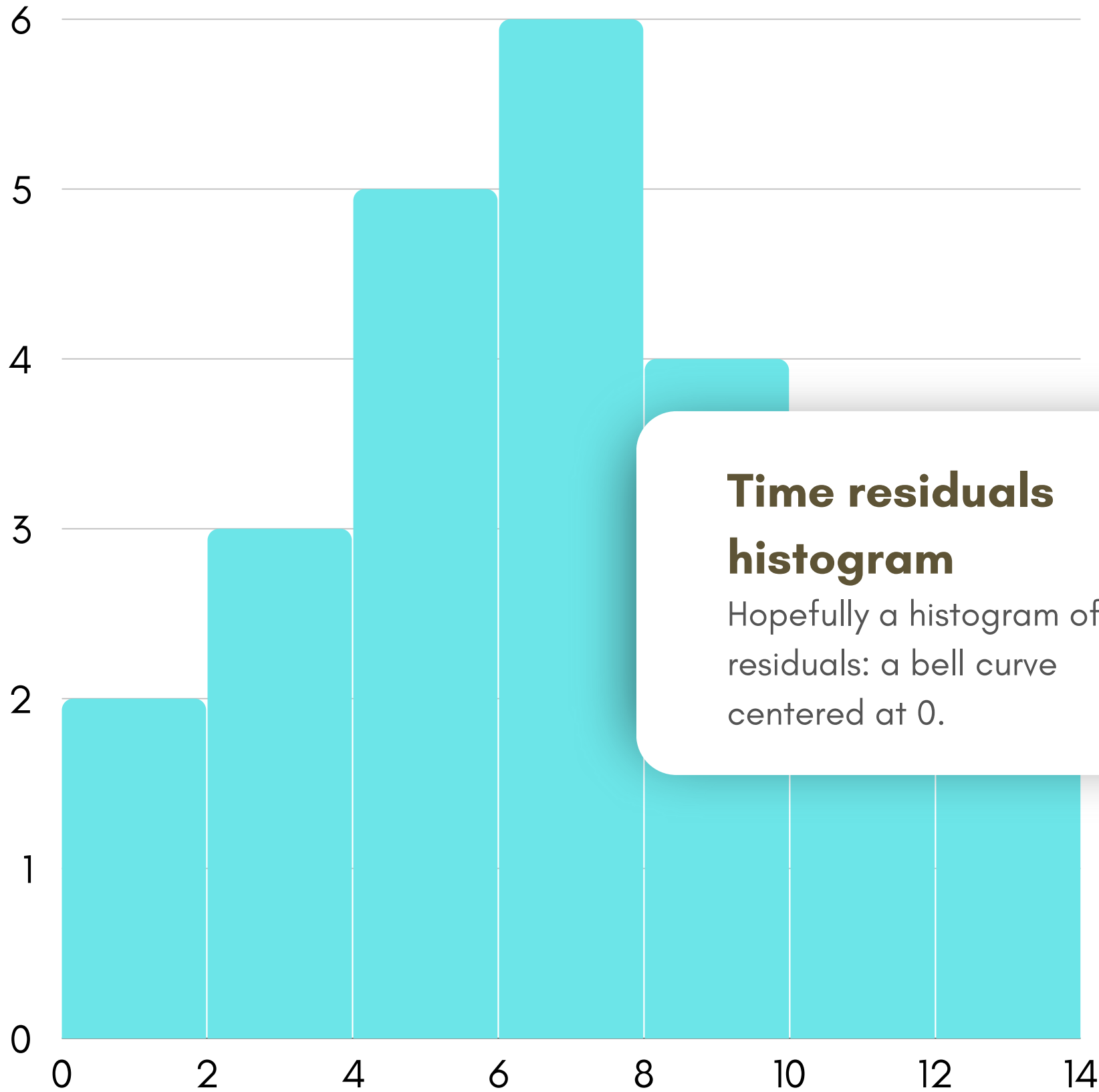
- Input: Raw 3-Component Waveforms (East, North, Vertical).
- Structure:
 - Encoder: Captures abstract features (frequency/shape).
 - Decoder: Restores original time resolution (<60s window).

- Critical Feature: Skip Connections transfer high-frequency details to ensure the timing remains accurate.

P:

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PERFORMANCE & VALIDATION

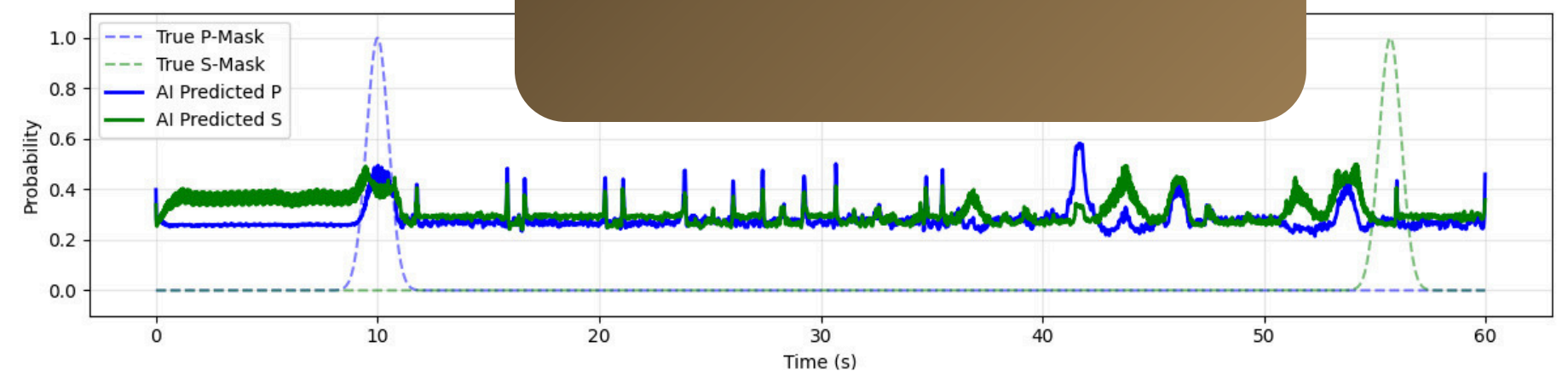


Time residuals histogram

Hopefully a histogram of time residuals: a bell curve centered at 0.

Classification Metrics

- Precision, Recall, F1-Score (Event vs. Noise), Confusion Matrix.





Thanks for Your Attention

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